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All-terrain vehicle

Abstract

An ATV is shown having a steering system comprised of a power steering unit having drive and driven pitman arms coupled to a drag link. The drive pitman arm is laterally offset from a vehicle centerline and is driven by the power steering unit. An alternate power steering system includes a pack and pinion subassembly coupled to a power steering motor, which then couples to steering arms of the ATV.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/062,105, filed Oct. 2, 2020, which is a continuation of U.S. patent application Ser. No. 15/895,495, filed Feb. 13, 2018, the entire disclosures of which are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

(1) The present disclosure generally relates to an ALL TERRAIN VEHICLE (ATV) and, more particularly, to an ATV having improved ergonomics and performance.

(2) Generally, ATVs are used to carry one or two passengers and a small amount of cargo over a variety of terrains. Due to increasing recreational interest in ATVs, specialty ATVs, such as those used for trail riding, racing, and cargo hauling have entered the market place. Most ATVs include an engine including between one and three cylinders. Most ATVs include a straddle or saddle type seat positioned above the engine.

(3) In one example of the invention, a method of making multiple widths of all-wheel drive vehicles is described which includes the steps of providing a frame having a front portion and a rear portion; providing a power train and coupling the power train to the frame, the power train comprising a primary motive unit, a transmission, a front final drive, a rear final drive and couplings between the transmission and the front and rear final drives; providing a plurality of front and rear suspension components of different lengths; providing a plurality of front and rear half shafts of different lengths; selecting sets of front and rear suspension components and front and rear half shafts to define a measured distance from a longitudinal centerline of the vehicle; and providing front and rear wheels, where a distance measured from outside the front and rear wheels is different for the different sets.

(4) In another illustrative embodiment, an all-terrain vehicle (ATV) comprises a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; front wheel steering mechanisms including right and left knuckles pivotally coupled to the respective front right and left suspension; a power steering unit coupled to the frame, the power steering unit including an output shaft which is laterally offset from a vertical centerline of the vehicle; right and left steering arms coupled to respective right and left knuckles, and a linkage coupling the output shaft to the steering arms, where the right and left steering arms are substantially the same length.

(5) In another illustrative embodiment, an all-terrain vehicle (ATV) comprises a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; front wheel steering mechanisms including right and left knuckles pivotally coupled to the respective front right and left suspension; a first stop member positioned on each of the front right and left suspension; and a second stop member positioned on each of the right and left knuckles; wherein the first and second stop members cooperate to provide extreme rotational positions.

(6) In another embodiment includes an all-terrain vehicle (ATV) comprising a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; a power steering unit coupled to the frame; a steering post providing torsional input to the power steering unit; a steering system support bracket coupled to the frame for supporting the power steering unit; and a steering stop coupled between the steering shaft and the steering system support bracket, to provide extreme rotational positions of the steering shaft.

(7) Finally, in another embodiment, an all-terrain vehicle (ATV) comprises a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions; a power steering unit coupled to the frame, the power steering unit comprising a power steering motor having an output shaft and a rack and pinion subassembly coupled to the power steering unit, the rack and pinion subassembly having a housing and a pinion gear driving coupled by the output shaft, and a rack drivingly coupled to the pinion gear; a steering

post providing torsional input to the power steering unit; and steering arms coupled to the rack; wherein torsional input to the power steering unit causes rotation of the output shaft and pinion, and linear movement of the rack and steering arms.

Description

- (1) The invention will now be described by way of reference to the drawing figures, where:
- (2) FIG. 1 is a front left perspective view of the ATV of the present disclosure;
- (3) FIG. 2 is a right rear perspective view of the ATV of FIG. 1;
- (4) FIG. 3 is a left side view of the ATV;
- (5) FIG. 4 is a right side view of the ATV;
- (6) FIG. 5 is a top view of the ATV;
- (7) FIG. 6 is a front view of the ATV;
- (8) FIG. 7 is a rear view of the ATV;
- (9) FIG. 8 is a front left perspective view of the frame for the ATV of FIGS. 1-7;
- (10) FIG. 9 is a right rear perspective view of the frame of FIG. 8;
- (11) FIG. 10 is a front left perspective view of a 55" wide version of the ATV showing the front suspension and steering system;
- (12) FIG. 11 is a front view of the ATV front suspension and steering system of FIG. 10;
- (13) FIG. 12 is an enlarged rear view of the right-hand suspension system of FIG. 11;
- (14) FIG. 13 is a front left perspective view of the wheel hub shown in FIG. 12;
- (15) FIG. 14 is an exploded view of the wheel hub coupling to the lower A-arm of FIG. 13;
- (16) FIG. 15 is a right rear perspective view of the ATV shown in FIG. 10;
- (17) FIG. 16 is an underside perspective view of the suspension and steering system;
- (18) FIG. 17 is an underside perspective view showing the electric power steering unit mounting and the steering arm linkage;
- (19) FIG. 18 is an exploded view of the electric power steering unit mounting and steering linkage of FIG. 17;
- (20) FIG. 19 shows a rear right perspective view of the steering linkage of FIG. 18;
- (21) FIG. 20 is a rear left perspective view of the linkage of FIG. 19;
- (22) FIG. 21 is a lower perspective view showing the driven pitman arm together with the steering stop;
- (23) FIG. 22 is an underside view showing the suspension of a 55" wide ATV;
- (24) FIG. 23 shows a view similar to that of FIG. 22 showing the suspension of a 48" wide ATV;
- (25) FIG. 24 shows a bottom view of the left side suspension of a 55" wide ATV;
- (26) FIG. 25 shows a bottom view of the left side suspension of a 48" wide ATV;
- (27) FIG. 26 shows a rear view of the rear suspension of a 55" wide ATV;
- (28) FIG. 27 shows a bottom view of the suspension of FIG. 24;
- (29) FIG. 28 shows a rear view of the 48" rear suspension;
- (30) FIG. 29 shows a bottom view of the suspension of FIG. 26;
- (31) FIG. 30 shows a right rear perspective view of the right rear wheel hub;
- (32) FIG. 31 shows a left rear perspective view of the mounting of the rear final drive and the frame of the ATV;
- (33) FIG. 32 shows a right rear perspective view showing the engine mounting together with the exhaust routing for the ATV frame;
- (34) FIG. 33 shows a front left perspective view of the ATV muffler of FIG. 32;
- (35) FIG. 34 shows a right rear view of the muffler of FIG. 31;
- (36) FIG. 35 shows a rear left perspective view of the hitch mount on the ATV;
- (37) FIG. 36 is a front view of an alternate rack and pinion steering system;

(38) FIG. 37 is a left hand side view of the rack and pinion steering system of FIG. 36;
(39) FIG. 38 is a bottom view of the rack and pinion steering system of FIG. 36;
(40) FIG. 39 is a left rear view of a riser block assembly usable with the handle bars; and
(41) FIG. 40 is an exploded view of the assembly of FIG. 39.
(42) Corresponding reference characters indicate corresponding parts throughout the several views. Unless stated otherwise the drawings are proportional.

DETAILED DESCRIPTION OF THE DRAWINGS

(43) With reference first to FIGS. 1-7, the ATV of the present disclosure will be described in greater detail. As shown, ATV is shown generally at 2 and comprises front wheels 4 and rear wheels 6. Front wheels 4 include rims 8 and tires 10, whereas rear wheels 6 include rims 12 and tires 14. Wheels 4 and 6 support a frame 20 which in turn supports a front body portion 22, a rear body portion 24 including a rack 26, a seat 28 and a mid-body portion 30 including a foot well 32. ATV 2 includes a steering system 36 including handlebars 38 for steering ATV 2, as described herein. A pod 40 is coupled to handlebars 38 and includes a headlight 42 (FIG. 1) and a control display 44 (FIG. 2). ATV 2 further comprises a powertrain 50 (FIG. 3) including an exhaust system 52 (FIG. 2). Finally, ATV 2 includes a front suspension system 56 (FIG. 6) and a rear suspension 58 (FIG. 7).

(44) As will be disclosed herein, ATV 2 can be provided in two widths, that is a 48" width and a 55" width. In accordance with the present disclosure, the frame 20, body components 22, 24 and 30, and powertrain 50 are identical for the two vehicle widths. Rather, the front and rear suspensions 56, 58 interchange to provide the wider vehicle, as disclosed herein.

(45) With reference now to FIGS. 8 and 9, frame 20 will be described in greater detail. As shown, frame 20 includes a front frame portion 70, a rear frame portion 72, a foot support portion 74 and a seat support portion 76. Frame 20 is generally defined by longitudinally extending tubes 80 having horizontal sections 80a, which turn upwardly and forwardly at portions 80b and turn upwardly and rearwardly at portions 80c. A removable forging 82 is provided on the driver's side of the vehicle, such that removal provides an access opening for installing or servicing components of the ATV. Foot support portion 74 is defined by a U-shaped tube 84 coupled to the two portions 80a and including mounting brackets 86. Seat support portion 76 is defined by longitudinally extending tubes 90 which couple at the top ends of frame tube portions 80b and 80c. Front ends of tubes 90 terminate in a cross brace 94, while tubes 90 support a rear cross brace 96.

(46) Front frame portion 70 includes front tubes 96 extending upwardly from a cross brace 98 to couple to a front side of cross brace 94. Upright tubes 100 extend upwardly from cross brace 98 and couple with tubes 96. Tubes 102 extend longitudinally between tube portions 80b and tubes 100, and support a steering system support bracket 110, as described further herein. Bracket 110 includes a lower bracket portion 112 fixed to tubes 102, for example, by welding. Frame 20 further includes lower tubes 114 which couple between tubes 80 and a front brace 116, where brace 116 includes an upright portion 118 which couples to cross brace 98. Four attachment brackets are provided for mounting of the front suspension 56, namely, a front lower bracket 126, a lower rear bracket 128, an upper front bracket 130 and an upper rear bracket 132. These same brackets 126, 128, 130 and 132 are replicated on the right hand side in a mirror image configuration.

(47) Rear frame portion 72 includes lower tubes 140 coupled to tubes 80a which extend rearwardly, while longitudinal tubes 142 extend rearwardly from a cross tube 146. Vertical tubes 148 extend upwardly between tubes 142 and 90. Braces 150 extend upwardly between tubes 140 and 142 and provide a first attachment point 156 (FIG. 9) for a rear lower coupling for the rear suspension 58 and an upper coupling point 160 for the rear suspension 58. A front lower bracket 166 is coupled to each of the tubes 140 for a front lower coupling point of rear suspension 58, while an upper bracket 170 provides coupling for a front upper coupling point for rear suspension 58. A muffler support 172 is positioned at a rear of the frame 20 and straddles frame members 142, and includes apertures 174.

(48) Finally and with respect to seat support portion **76**, a forging **180** is provided, interrupting a length of the passenger side tube **90** which couples to cross braces **182** and which is removable for the assembly and/or servicing of the vehicle components, such as powertrain components.

(49) With reference now to FIGS. **10-16**, the front suspension **56** of the 55" vehicle will be described. With reference first to FIG. **10**, a front end of the vehicle is shown without the wheels and the front body **22** which exposes the front suspension **56** and frame **20**. A front drive **190** is also shown to drive the front wheels **4** from the powertrain. As is known, at the front end of the vehicle, the steering system **36**, the front suspension **56** and the front drive **190** all cooperate together.

(50) As shown in FIG. **10**, front suspension **56** includes a lower left A-arm **200**, upper left A-arm **202**, lower right A-arm **204** and upper right A-arm **206**. Shock absorbers **208** and **210** couple between upper A-arms **202**, **206** and upper brace **94**, respectively. Inner ends of lower left A-arm **200** couple to coupling points **126**, **128** (FIG. **8**), while inner ends of upper left A-arm **202** couple to coupling points **130**, **132** (FIG. **8**). Right A-arms **204**, **206** are likewise coupled to like coupling points of the frame **20** on the right-hand side. As shown best in FIG. **12**, upper A-arms **202** and **206** have an upwardly directed bend at **220** which is profiled to provide clearance for steering arm **222**. As will be discussed further herein, steering arm **222** is coupled at an inner end to a motor of the steering system **36**, and is coupled at its outer end to knuckle **224**. Knuckle **224** is coupled to A-arms **204**, **206** by way of pins **230**, **232**. Knuckle **224** includes an extension **234** which couples to threaded posts **236** of steering arm **222**. Wheel hubs **240** are provided having studs **242** for mounting wheels **4**.

(51) As shown best in FIG. **13**, hub **240** includes a constant velocity joint **250** having an inner profiled geometry **252** which is suitable for driving by way of a half shaft (or constant velocity (CV) shaft) **256** (FIG. **10**), as is known in the art. Inner ends of half shafts **256** likewise include drive portions to be driven by constant velocity couplings **260** (FIG. **10**) from the drive unit **190**. As shown best in FIGS. **13** and **14**, knuckle **224** includes an integrated stop member **266** in the form of an arcuate slot having extreme contact surfaces **268** and **270** (FIG. **14**). Meanwhile, lower right A-arm **204** includes a coupling **276** having an opening at **278** which is received over pin **230** to receive a castellated nut **280** (FIG. **13**) to couple lower A-arm **204** to knuckle **224**. As shown, coupling **276** includes an integrated stop member **284** defining an upstanding post having a shark fin geometry, whereby hub **224** can rotate about an axis defined by pins **230**, **232** to the extreme positions where surface **268** or **270** contacts stop member **284**.

(52) The forging also makes contact at a 45° angle instead which increased the strength significantly. This design is lighter and gives greater control to the stop. The shark fin **284** is wider at the base **284a**, as compared to the top **284b**, (see FIG. **14**) which allows a progressive steering stop and limits the steering angle as the wheel travels downward to protect the half shafts **256** (FIG. **10**) at suspension full extension.

(53) With reference now to FIGS. **15-21**, steering system **36** will be described in greater detail. With reference first to FIG. **15**, steering system **36** includes a steering post **300** extending downwardly from handlebars **38** (FIG. **1**) to a power steering motor **302**. As shown best in FIG. **17**, steering system support bracket **110** further comprises an upper bracket portion **304**. Steering motor **302** is coupled to bracket portion **304** (FIG. **16**), which in turn couples to lower bracket portion **112** (FIG. **15**). With reference now to FIGS. **17** and **18**, the mounting of steering motor **302** will be described in greater detail. As shown best in FIG. **18**, power steering motor **302** includes a motor portion **310** and an output portion **312** having a splined drive shaft **314** extending through a bearing cap **316**. Four threaded bosses **318** are provided for coupling to bracket **304**. As shown, bracket **304** includes an upper plate **320** having a bracket tab **322** extending from one end thereof having apertures **324** and **326**. At the opposite end, a bracket sidewall is provided at **330** having apertures at **332** (FIG. **17**). Plate portion **320** includes an enlarged aperture at **336** (FIG. **19**) which fits over bearing cap **316**, as best shown in FIG. **17**. Plate portion **320** further includes apertures

340 (FIG. 19) which align with the threaded bosses **318** to receive fasteners **344** therethrough coupling bracket **304** to a lower side of steering motor **302**, as shown best in FIG. 18.

(54) As should be appreciated, when bracket **304** is coupled to the bottom of steering motor **302**, splined shaft **314** protrudes through aperture **336**. A drive pitman arm **350** is provided having a coupling **352** having an inner spline **356** (FIG. 19). Pitman arm **350** includes an arm portion **360** having an aperture **362** adjacent a free end of the arm **360** and an aperture **364** intermediate aperture **362** and shaft portion **370**. Shaft portion **370** is received in a bearing assembly **376** having a plate portion **378** (FIG. 21) including a rotatable stop portion **380** having stop surfaces **380a** and **380b** with a bearing **384** extending therethrough. Bearing assembly **376** can be coupled to lower bracket **110** as described herein. Lower bracket **112** includes a plate portion **400** (FIG. 18) having an aperture at **402** to receive bearing portion **384**. Plate portion **400** includes square apertures **404** which align with square apertures **406** on bearing assembly **376** which, as can be appreciated, receive carriage bolts therethrough to couple the bearing assembly **376** to the plate portion **400**.

(55) As shown best in FIG. 21, drive pitman arm **350** includes a stop member **410** which rotates between the extreme portions of surfaces **380a** and **380b** under the influence of splined shaft **314**. Finally, drive pitman arm **350** includes a groove **416** (FIG. 21) to receive a snap ring **418** (FIG. 18), therein which retains the shaft portion **370** in the bearing **384**, and the pitman arm **350** to the splined shaft **314**. The steering stop **380**, **410** on the post protect the CV shafts **256** primarily, the post stop protects the wheels **4** from contacting the bodywork during neutral suspension position. The plate **380** is interchangeable and allows adjustment of the stop position which is important at the neutral position when larger tires are desired.

(56) As shown best in FIG. 18, a driven pitman arm is shown at **420** which is similarly constructed as pitman arm **350**, however it does not have a splined shaft as it is driven by pitman arm **350**. Rather, pitman arm **420** includes a cylindrical bearing portion **422** having a through opening at **424**. Pitman arm **420** also includes an arm portion **428** having apertures **430** and **432**. As shown best in FIG. 18, bracket **112** includes a portion **440** (FIGS. 18-19), which is coupled to tubes **102** having upper plate portion **442** having an aperture **444** aligned with aperture **446** of lower plate portion **400**. Thus, as shown in FIGS. 19, 20, driven pitman arm **420** can be coupled between bracket portions **112**, **304** by way of a fastener **450** extending through aperture **326** of tab portion **322** extending downwardly through aperture **424** of driven pitman arm **420** and through apertures **444**, **446** of bracket portion **112**. A complementary fastener **452** may be coupled to fastener **450** to retain driven pitman arm **420** in place.

(57) With reference still to FIGS. 18 and 19, drive and driven pitman arms **350**, **420** are coupled together by way of a drag link **460**, where drag link **460** includes a bar portion **462** having couplings **464** at each end. Each coupling **464** includes an aperture **466** which align with apertures **362** and **430** to receive fasteners **470** therethrough. Fasteners **470** couple with fasteners **472** to retain drag link **460** in a coupling relationship with the drive pitman arm **350** and driven pitman arm **420**, as shown best in FIG. 19.

(58) Finally, with respect to FIG. 18, steering arms **222** each include a ball joint **500** having a threaded end **502**, where each of the threaded ends **502** is received in one of the apertures **364** or **432**. Thus, as should be appreciated, input from the splined shaft **314** to drive pitman arm **350** causes a like rotation of the driven pitman arm **420** by way of drag link **460**, which in turn causes transverse movement of the steering arms **222** in a push-pull relationship with the pitman arms **350**, **420** depending upon the direction of rotation.

(59) With reference now to FIGS. 11 and 16, further features of the steering system will be described. As shown best in FIG. 11, the steering shaft **300** extends along a non-vertical axis **510** with respect to vertical axis **512** to provide an angular offset $\beta_{\text{sub.1}}$, where $\beta_{\text{sub.1}}$ =approximately 8.5°, but could be in a range from 5°-12°, and more particularly from 7°-10°. This positions the center of the left inner ball joint **500** (FIG. 16) at a left vertical position **516**. As shown in FIG. 11, vertical position **516** is offset from the vertical center line **520** and the vertical center **522** of the

right inner ball joint **500** is laterally offset from vertical center line **520** as well. As shown, left inner ball joint **500** is offset from the vertical center line **520** by a distance of **526**, whereas the right inner ball joint **500** is offset from the vertical center line **520** by a distance of **528**, where distances **526** and **528** are equal, and are equal to 3.25", but could be in a range from 2"-5. Thus, the laterally offset steering motor in combination with the drag link **460** eliminates the bump steer that is introduced into the suspension.

(60) Furthermore, with reference to FIGS. **16** and **19**, each of the arm portions **360** and **428** of respected pitman arms **350** and **420** include portions skewed at **360a** and **428a** relative to a horizontal axis **530** such that portions **462a** and **462b** of drag link **460** form angles $\beta_{sub.2}$ relative to horizontal axis **530**, where $\beta_{sub.2}=15^\circ$, but could be in range from 10° - 20° .

(61) With reference now to FIGS. **22** and **23**, the front suspension of the 48" and 55" vehicles will be distinguished. With reference first to FIG. **22**, ATV **2** is in the form of the 55" vehicle where a distance from outer faces of wheel hubs **240** is shown by a lateral distance $X_{sub.1}$, where $X_{sub.1}=49"$. In this way, the distance $X_{sub.1}$ plus the width dimensions of the tires would approximate the 55" width vehicle. It should also be understood that the suspension as described with respect to FIGS. **10-16**, depict the 55" vehicle such that the upper and lower A-arms **200**, **202**; **204**, **206** relate to the 55" vehicle. The half shafts **256** are also of a different length to accommodate the wider vehicle. As shown in FIG. **22**, the wheel hubs **240** are positioned further forward relative to the frame **20** such that the half shafts **256** sweep forward relative to a horizontal center line **540** through the couplings **260**. As shown, the half shafts **256** are swept forward by an angle $\beta_{sub.3}$, where $\beta_{sub.3}=2.8^\circ$, but could be in a range from 0° - 5° .

(62) With reference now to FIG. **23**, the front suspension for the 48" vehicle is shown at **56'**, where the front suspension **56'** includes left A-arms **200'** and **202'** and right A-arms **204'** and **206'**. The half shafts **256'** are also of a different length to accommodate the lateral difference between the couplings **260** and the hubs **240**. As shown, the lateral distance between front faces of the hubs **240** is $X_{sub.2}$, such that the distance $X_{sub.2}$ plus the width of the wheels equates to the 48" vehicle. As shown, $X_{sub.2}=42"$. As also shown in FIG. **23**, the half shafts **256'** are swept rearwardly relative to the lateral center line **540** such that an angle $\beta_{sub.4}$ is defined which reduces the wheel base of the vehicle relative to the 55" vehicle described in FIG. **22**. As shown, the steering system **36** of the 48" and 55" vehicle are identical with the exception that the steering arms **22'** of the 48" vehicle are somewhat shorter to accommodate the width difference of the vehicles. As also shown, $\beta_{sub.4}=7.4^\circ$, but could be in a range from but could be in a range from 5° - 10° .

(63) With reference now to FIGS. **24** and **25**, the front suspension of the 48" and 55" vehicles will be described. With reference first to FIG. **24**, the 55 inch vehicle is shown where the front left wheel is turned to the maximum left position, to a rotation of $\beta_{sub.5}=35^\circ$. Due to the fact that the half shafts **256** sweep forward on the 55" vehicle (FIG. **22**) the first contact between the CV joint **250** (FIG. **13**) and the half shaft **256**, would be at the rearward side of the CV joint **250**. Thus the upstanding post **284** on the 55" vehicle contacts the rear contact surface **270** to form the extreme position for the 55" vehicle.

(64) With reference now to FIG. **25**, the 48 inch vehicle is shown where the front left wheel is turned to the maximum right position, to a rotation of $\beta_{sub.6}=35^\circ$. Due to the fact that the half shafts **256'** sweep rearward on the 48" vehicle (FIG. **23**) the first contact between the CV joint **250** (FIG. **13**) and the half shaft **256'**, would be at the forward side of the CV joint **250**. Thus the upstanding post **284** on the 48" vehicle contacts the front contact surface **268** to form the extreme position for the 48" vehicle.

(65) It should also be understood that the knuckles **224**, hubs **240** and the couplings **276** are identical for the 48" and 55" vehicles and that the stops **268**, **270** are defined for providing the extreme position for both vehicles.

(66) With reference now to FIGS. **26-31**, the rear suspension **58** of the ATV **2** will be described. With reference first to FIGS. **24** and **25**, the suspension of the 55" vehicle will be described. As

shown, the suspension **58** is comprised of a left lower A-arm **550**, a left upper A-arm **552**, a lower right A-arm **554** and a right upper A-arm **556**. The left A-arms **550**, **552** are coupled to a knuckle **560** which retains a left rear wheel hub **562**. Wheel hub **562** is driven by a rear left half shaft **564**. The right A-arms **554** and **556** are coupled to a right rear knuckle **570** and a right rear wheel hub **572** where wheel hub **572** is driven by a right rear half shaft **574**. A torsion bar **580** couples the lower A-arms **550**, **554** by way of vertical posts **582**. As shown, half shafts **564**, **574** are driven by couplings **586** of a final drive **590**.

(67) With reference now to FIGS. **28** and **29**, the rear suspension **58'** for the 48" vehicle will be described. As with the front suspension, the majority of the components of the vehicle remain the same, but the left A-arms **550'**, **552'** and half shaft **564'** are replaced to compensate for the vehicle width. Similarly, the right-hand side suspension is also changed with lower A-arm **554'**, upper A-arm **556'** and half shaft **574'** having the reduced dimensions like the left-hand side. The torsion bar **580'** is also different to accommodate the width of the vehicle.

(68) With reference now to FIG. **28**, the coupling between the A-arms and the knuckle will be described. As shown in FIG. **28**, the right rear suspension is shown where A-arms **554** and **556** are coupled to knuckle **570**. This is also the same knuckle and wheel hub used on the 48" vehicle and mirror images of the left-hand side wheel hub. The knuckle **570** includes a single elongate coupling **590** at the upper end thereof for coupling to upper A-arm **556** and includes two lower couplings at **592** for attachment of A-arm couplings **594**. Wheel hub **572** includes a constant velocity joint **600** extending through the knuckle **570** which includes a drive profile **602** for matching with a constant velocity joint on half shaft **574**. As shown in FIG. **28**, the coupling points **592** are raised relative to the constant velocity coupling **600** such that the top of the coupling points **594** are higher than the inner geometry **602**.

(69) With reference now to FIG. **31**, the rear final drive **590** is shown mounted in the rear frame portion **72** positioned above frame tubes **140** and below frame tubes **142**. Rear suspension mounts **150**, **166** and **170** are shown in greater detail. Final drive **590** is shown positioned adjacent to forging **82** which may be removed to access the final drive **590**.

(70) With respect now to FIGS. **32-35**, the exhaust system **52** will be described in greater detail. As shown, exhaust system **52** includes an exhaust pipe **610** coupled to engine **612** at a front end thereof, and extending rearwardly past continuously variable transmission (CVT) **614** to an elbow **616**, and then to a front end of muffler **620**. As shown, muffler is centered in the vehicle frame **20** above tube **146**, intermediate tubes **80c** and below tubes **90**. As shown best in FIG. **32**, muffler **620** includes a muffler mounting post **630**, having post portions **632** extending into apertures **174** (FIG. **35**) of muffler mount **172**. The muffler **620** is centrally mounted because with the extended travel of suspension the muffler **620** cannot be packaged in the wheel well at full rebound/jounce.

(71) It should be understood that the vehicle **2** includes front and rear drive shafts coupled between U-joint **634** of front drive **190** (FIG. **15**) and output U-joint **635** (FIG. **32**); and between U-joint **636** of rear drive **590** (FIG. **31**) and output U-joint **638** (FIG. **32**). The drive shafts are similar to those described in U.S. Pat. No. 9,873,316, the subject matter of which is incorporated herein by reference.

(72) With reference now to FIGS. **36-38**, an alternate steering system is shown as a rack and pinion system **640** which is an alternative to the steering system shown in FIG. **18**. Rack and pinion system **640** utilizes the power steering motor **302** as disclosed in the embodiment of FIG. **18**, but couples to it a rack and pinion subassembly **642**. As shown, subassembly **642** includes a housing **644** having bearing covers **646** and **648** housing bearings **650**, **652**. Bearings **650** and **652** support a stub shaft **656** which includes a splined internal shaft which couples to splined drive shaft **314** of power steering motor **310**. A pinion **660** is coupled to stub shaft **656**, and as best shown in FIG. **38**, pinion **660** has an arcuate shape at **662** having gear teeth which mesh with teeth **664** of a rack **666**. The rack includes couplers **668** which couple to steering arms **222** (FIG. **18**) as opposed to the utilization of pitman arms **350**, **420**. Thus, as shown in FIG. **38**, when pinion **660** rotates in a

clockwise direction, rack **666** moves linearly in the direction of arrow **670** and when pinion **660** rotates in a counterclockwise direction, rack **666** moves linearly in the direction of arrow **672**.

(73) With reference now to FIGS. **39** and **40**, a steering riser block assembly **680** and an adjustable instrument cluster **682** will be described herein. As shown, riser block assembly **680** is coupled to an alternate steering arm **684** having a T-shaped post at **686**. Riser block assembly **680** includes the main body portion **690**, which in the disclosed embodiment is an aluminum block extruded in the direction of apertures **692** in post **694**. Block **690** includes an upper surface **696** and a lower surface **698** which receives upper clamps **700** and lower clamps **702** respectively. Lower clamps **702** include an arcuate opening at **706** for placement against T-shaped post **686** whereby fasteners **708**, **710** may be utilized to attach clamps **702** to bottom surface **698** trapping T-shaped post **686** therebetween. Lower end of main body portion **690** includes an arcuate opening at **712** to receive the T-shaped post **686**. It should be appreciated that a lower end of aperture **692** is threaded, to receive threads on fasteners **708**.

(74) As shown best in FIG. **40**, instrument cluster **682** includes a bracket **716** including a lower plate portion **718** having apertures **720**. Apertures **720** align with apertures **722** of upper clamps **700**. Thus, fasteners **726** may be received through apertures **720** of bracket **716**, through apertures **722** of clamps **700** into the upper end of apertures **692**. It should be appreciated that the upper ends of apertures **692** are also threaded to receive threads on fasteners **726**. Thus, when bracket **716** is coupled to riser block **690**, handlebar **740** is coupled therebetween and upper clamps **700** and riser block **690** grips a knurled surface **742** to retain the position of the handlebars **740**. Upper end on main housing **690** includes an arcuate opening **744** and clamp **700** includes an arcuate opening **745** to grip knurled surface **742**. Bracket **716** includes a plurality of bracket arms **746**, **748**, **750** which allows coupling of upper housing **760** of pod **682** by way of fasteners **762** to bracket **716**.

(75) Thus, given the geometry of the riser block **690**, the riser block **690** may rotate towards and away from the rider on T-shaped post **686** which moves handlebars **740** towards or away from the rider to accommodate the ergonomic feel for different sized riders. This also adjusts the pod **682** to accommodate different sized riders. Furthermore, the riser block **690** allows the handlebars **740** to be rotated downwardly in a stored or shipped position as shown in the phantom lines of FIG. **39**.

(76) While this invention has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

Claims

1. An all-terrain vehicle (ATV), comprising: a frame; ground engaging members supporting the frame, including front and rear ground engaging members; front right and left suspensions, at least one of the right and left suspensions including an upper arm and a lower arm; front wheel steering mechanisms including right and left knuckles pivotally coupled to the respective front right and left suspensions; a power steering unit coupled to the frame, the power steering unit including an output shaft which is coupled to the right and left knuckles for steering; a first stop member positioned on each of the front right and left suspension, the first stop member defined on the lower arm; and a second stop member positioned on each of the right and left knuckles and configured to receive a portion of the first stop member therein, wherein the first and second stop members cooperate to provide extreme rotational positions for the power steering unit, wherein the extreme rotational positions are defined by the knuckle.
2. The all-terrain vehicle of claim 1, wherein the first stop member includes an upstanding post.
3. The all-terrain vehicle of claim 2, wherein the upstanding post has a shark fin geometry.
4. The all-terrain vehicle of claim 1, wherein the first stop member is configured to contact the

second stop member at a 45-degree angle.

5. The all-terrain vehicle of claim 1, wherein the second stop member defines an arcuate slot.

6. The all-terrain vehicle of claim 1, wherein the knuckle defines a pin and the lower arm defines an opening, wherein the pin is configured to be received within the opening to couple the lower arm to the knuckle.

7. The all-terrain vehicle of claim 6, wherein the lower arm is rotatable about the pin and the lower arm is configured to rotate between the extreme rotational positions.

8. A suspension structure of an all-terrain vehicle comprising: a shock absorber; an upper arm and a lower arm operatively coupled to the shock absorber; a knuckle coupled between the upper arm and the lower arm; a stopper defined by the knuckle and configured to regulate a displacement angle between the lower arm and the knuckle, the stopper defining an arcuate slot; and a projection defined by the lower arm and configured to abut surfaces of the stopper during rotation.

9. The suspension structure of claim 8, wherein the projection defines a top and a base opposite the top, wherein a width of the projection at the top is less than a width of the projection at the base.

10. The suspension structure of claim 9, wherein the top of the projection is configured to abut the surfaces of the stopper during rotation.

11. The suspension structure of claim 8, wherein the stopper includes a first end and a second end opposite the first end, and the surfaces of the stopper include a first extreme contact surface at the first end and a second extreme contact surface at the second end.

12. An all-terrain vehicle comprising: a frame; ground engaging members supporting the frame, including front and rear ground engaging members; a front suspension including an upper arm and a lower arm; a front wheel steering mechanism including a knuckle pivotably coupled to the front suspension; a power steering unit coupled to the frame and configured for steering, the power steering unit including an output shaft which is coupled to the knuckle; a projection member defined by the lower arm; and a stopper defined by the knuckle and configured to regulate a displacement angle between the lower arm and the knuckle, wherein the stopper defines an arcuate slot, and wherein the projection is configured to abut surfaces of the stopper during rotation of the power steering unit.

13. The all-terrain vehicle of claim 12, wherein the projection defines a top and a base opposite the top, wherein a width of the projection at the top is less than a width of the projection at the base.

14. The all-terrain vehicle of claim 12, wherein the knuckle is operatively coupled to a wheel hub of one of the front ground engaging members.

15. The all-terrain vehicle of claim 12, further including a steering arm coupled to the knuckle and positioned between the upper arm and the lower arm.

16. The all-terrain vehicle of claim 15, wherein the knuckle defines a top end, a bottom end opposite the top end, and a length therebetween, and wherein the knuckle includes an extension defined along the length and the steering arm is coupled to the extension.

17. The all-terrain vehicle of claim 15, wherein the upper arm has a first end, a second end, and a length therebetween, and the upper arm defines an upwardly directed bend along a portion of the length configured to provide clearance to the steering arm.

18. The all-terrain vehicle of claim 12, wherein the lower arm has a first end, a second end, and a length therebetween, wherein the lower arm is curved along the length.

19. A suspension structure of an all-terrain vehicle comprising: a shock absorber; an upper arm and a lower arm operatively coupled to the shock absorber; a knuckle coupled between the upper arm and the lower arm; a stopper defined by the knuckle and configured to regulate a displacement angle between the lower arm and the knuckle, wherein the stopper includes a first end and a second end opposite the first end, and the surfaces of the stopper include a first extreme contact surface at the first end and a second extreme contact surface at the second end; and a projection defined by the lower arm and configured to abut surfaces of the stopper during rotation.
